MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE C1A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, inert gas

See figure 1

Mounting position: Any

Weight: 3-ounces nominal (85 grams)

ABSOLUTE RATINGS:

Parameter:	Ef	Epp	epx	epy	Ib	ib	i surge 0.1 sec	tk	TA	F
Unit:	V	Vac	v	v	Adc	a	aac	sec	°C	Hz
Maximum:	2.75	---	340	170	0.64	7.7	77	---	70	380
Minimum:	2.25	---	---	---	---	---	---	40	-55	---
TEST CONDITIONS:	2.5	110	---	---	0.64	---	---	---	---	60 (Note 1)

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current	-		0.65	II	If	5.5	6.5	Aac
3206	Operation	-	Rg = 10,000 ohms	0.65	II	Ec	+ 1.35	-2.75	Vdc
1231	Peak emission by voltage drop	10	Grid connected to anode through 10,000 ohms; ib = 7.7 a	0.65	II	etd	---	30	v
---	AC starting voltage	4	Ec = +3 Vdc; Rg = 1,000 ohms	0.65	II	Ep	---	45	Vac
(C) 3204	Voltage drop	5	Ec = +3 Vdc; Rg = 1,000 ohms Epp = 50 Vac; t = 300 (min)	0.65	II	Etd	---	11.5	Vdc
---	Peak forward voltage	6	Rg = 1,000 ohms	0.65	II	Ec	---	-5.3	Vdc
3216	Grid current	7	Rg = 10,000 ohms and 1 Meg	0.65	II	Ic	---	5	µAdc
	Quality conformance inspection, part 2								
---	No load operation	2,8	Ep = 0; t = 100 hours	---	---	---	---	---	---
	Quality conformance inspection, part 3								
---	Life test	-	Group D; operation; t = 500 hours	---	---	---	---	---	---
---	Life-test end points;								
3206	Operation	-		---	---	Ec	+ 1.35	-2.75	Vdc
3216	Grid current	-		---	---	Ic	---	5	µAdc
1231	Peak emission by voltage drop	-		---	---	etd	---	45	v
1031	Low-frequency vibration	2,9		---	---	---	---	---	---
1041	Shock	2,9	Hammer-angle = 24°; Ef = 2.5 V ± 10%	---	---	---	---	---	---
---	Cold starting	2,3, 9	Grid to anode through 10,000 ohms	---	---	---	---	---	---

NOTES:

1. The following circuit conditions shall apply for all tests.

   (a) Anode and grid circuit returns to center tap of secondary of filament transformer.
   (b) Filament transformer terminals phased so that filament prong No. 1 is positive relative to filament prong No. 4 when anode is positive.

2. After completion of this test, operate tube for 5 minutes at optional Epp and Ib = 0.64 Adc and then test for heater current, operation, peak emission, ac starting voltage, voltage drop, peak forward voltage, and grid current.

3. Apply rated filament and anode voltage simultaneously for a 3-minute period with the load at 0.1 ampere dc and then allow the tube to cool for 15 minutes. This application shall be repeated 500 times.

4. The anode voltage shall be lowered until the tube cuts off. The anode voltage, at the instant after the tube cuts off, shall be within the limit specified.

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5. With an ac voltage source, and a resistance load, measure the power loss in the tube with a wattmeter connected so that the current coil is in series with, and the voltage coil across, the tube and current coil. The tube voltage drop (Etd) is the total power loss measured above minus the current coil loss under load divided by the average or dc current (Ib).

6. The anode voltage shall be increased to 120 Vac, with the anode current held at a value not greater than 0.64 Adc. The grid bias shall be increased in a negative direction until the tube cuts off and shall be within the limit specified.

7. Operate tube at least 5 minutes prior to test. Operation may be at optional Epp but Ib must be 0.64 Adc minimum. To test, switch to test voltage and Ib = 0.64 Adc in less than 2 seconds.

8. The peak forward voltage is measured before and after this test with Ec = -10 Vdc; Rg = 10,000 ohms. The maximum value shall not shift more than 100 volts from the initial reading after the 100-hour no-load operation period.

(C) 9. This test shall be performed annually. A sample size of three tubes shall be used, with no failures allowed. In case of a sample failure, the test shall become a quality conformance inspection (QCI), part 2 test, for three successful consecutive submissions, at which time the test may revert to the QCI, part 3 test.

10. This test is to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

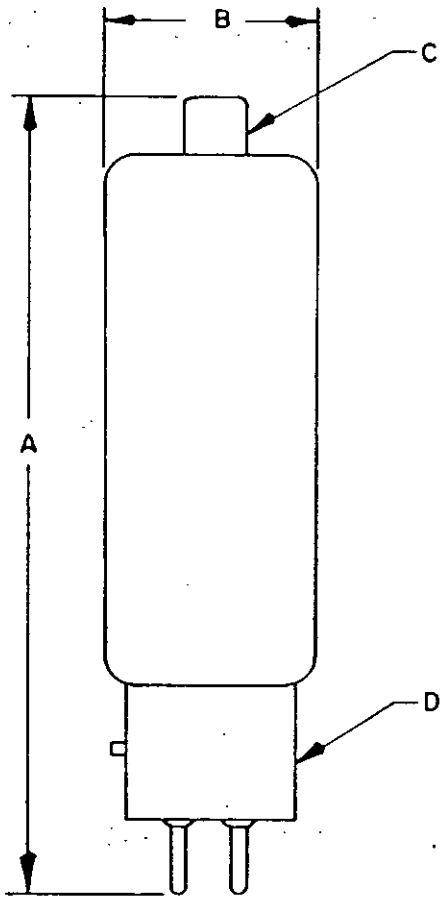
User activities:
Army - MU
Navy - AS, OS, MC, CG, SH

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-2869)

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Pin connections	
Pin no.	Element
1 & 4	f
2	nc
3	g
Cap	a

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	6.500 (165.10)	6.750 (171.45)
B	1.438 (36.53)	1.687 (42.85)
© Quality conformance inspection, part 3 (see note)		
C	Cap: C1-5 (EIA)	
D	Base: A4-10 (EIA)	

NOTE:
Dimensions listed under quality conformance inspection (QCI), part 3, shall be checked annually, using a sample of three tubes with no failures allowed. In case of a sample failure, the failing dimension(s) shall become a part of QCI, part 2, for three consecutive successful submissions, at which time the test may revert to the QCI, part 3 test.

FIGURE 1. Outline drawing of electron tube type C1A.

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